Pavel Perezhogin

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scholar.google.com/citations?user=HRXHqugAAAAJ

Education and Employment

Oct 2025 – Now — **Research Scientist** in Mathematics Department. Courant Institute School of Mathematics, Computing, and Data Science, New York University. **Lead Scientist** in M^{2}LInES project. Advisors: Prof. Laure Zanna, Prof. Alistair Adcroft, Prof. Carlos Fernandez-Granda

Dec 2021 – Oct 2025 — **Postdoctoral Associate**. Same Appointment.

Jun-Jul 2024 — **Short-term visit**. Stanford University, Center for Turbulence Research, leading a project in **Summer Program 2024**

2017 – 2021 — **PhD** in Mathematical Modeling, Numerical Methods and Software.
Marchuk Institute of Numerical Mathematics of the Russian Academy of Sciences.
Thesis (ru, en): Stochastic and deterministic subgrid parameterizations for two-dimensional turbulence and their application in ocean circulation models. Advisor: Dr. Andrey Glazunov

2011 – 2017 — **BSc&MSc** in Applied Mathematics and Physics. Department of Control and Applied Mathematics. Moscow Institute of Physics and Technology (MIPT).

Awards

2019 — **PhD Research Fellowship**, Russian Foundation for Basic Research, Grant No. 19-35-90023. Advisor: A. V. Glazunov.

2018 — **Medal of the Russian Academy of Sciences** for students for the best scientific work in oceanology, atmospheric physics and geography.

Contributions to ocean models

- My data-driven parameterizations of ocean mesoscale eddies were included in the numerical ocean model GFDL MOM6 (link1, link2, link3)
- My ocean parameterizations developed during the PhD were included in the numerical ocean model NEMO and evaluated in a recent PhD thesis at Sorbonne University (link)
- I developed an advanced HPC implementation with load balancing for the FEMAO ocean model (link)

Additional Experience

Leadership — Responsible for the ocean component of **CESM-MLe** (Community Earth System Model – Machine Learning Enhanced) as part of the Integration Team, dedicated to incorporating data-driven parameterizations and techniques into the CESM earth system model.

Teaching — **Invited** guest lectures "Machine Learning in Geophysics", Russia, Moscow, INM RAS (2023).

Mentoring — Grad. student Ivan Kobzar (co-advised with Dr. Andrey Glazunov), and undergrad. student Matias Ortiz, grad. student. Sid Arora, and postdoc Yuan Yuan (co-advised with Prof. Laure Zanna).

Reviewing — Journal of Advances in Modeling Earth Systems (JAMES) | Ocean Modeling | Geoscientific Model Development (GMD) | Physics of Fluids

Selected Talks and Posters

2026 — Mathematical Geosciences seminar at CU Boulder (**invited**), Machine Learning for Ocean modeling, UK (**invited, Keynote speaker**), Gordon Research Conference on Ocean Mixing, CESM Ocean Model Working Group

2025 — American Geophysical Union (AGU) annual meeting, Climate Process Team (CPT) annual meeting, NCAR-DLR meeting (Boulder, Colorado), CESM Ocean Model Working Group

2024 — Ocean Sciences | CESM Ocean Model Working Group

2023 — Courant Atmosphere Ocean Science Colloquium (**invited**) | AGU Fall Meeting | APS Division of Fluid Dynamics | CESM Workshop | CPT Annual Meeting | NEMO Machine Learning WG (**invited**) | CESM Ocean Model WG meeting

2022 — AGU Fall Meeting | CPT Annual Meeting | NEMO Eddy Closure WG (**invited**)

2017-2021 — TRR 181 Seminar (**2021, invited**) | EGU General Assembly (**2020, 2021, online**) | AGU Fall Meeting (**2020, online**) | ECMWF Annual seminar (**2020, online**) | Winter School in Les Houches (**2019**, **poster**) | 32nd-IUGG (**2018**) | CITES-**2017**

Selected Publications

Peer-reviewed Journal papers

1. Balwada, D., **Perezhogin**, **P.**, Adcroft, A., & Zanna, L. (2026). Design and implementation of a data-driven parameterization for mesoscale eddy-induced transport: Idealized two-layer configurations. *Journal of Advances in Modeling Earth Systems*, *18*(9), e2025MS005245. doi:https://doi.org/10.1029/2025MS005245
2. Kamm, D., **Perezhogin**, **P.**, Deshayes, J., Meunier, E., & Barge, A. (2026). Assessing data-driven eddy-parameterizations in an atlantic sector model. *Journal of Advances in Modeling Earth Systems*, *18*(6), e2026MS005816. doi:https://doi.org/10.1029/2026MS005816
3. **Perezhogin**, **P.**, Adcroft, A., & Zanna, L. (2026). Calibration of a neural network ocean closure for improved mean state and variability. *Geophysical Research Letters*, *53*(13), e2026GL123513. doi:https://doi.org/10.1029/2026GL123513
4. Wu, J., **Perezhogin**, **P.**, Gagne, D. J., Reichl, B. G., Subramanian, A. C., Thompson, E. J., & Zanna, L. (2026). Data-driven probabilistic air-sea flux parameterization. *Geophysical Research Letters*, *53*(6), e2025GL120472. doi:https://doi.org/10.1029/2025GL120472

5. **Perezhogin**, **P.**, Adcroft, A., & Zanna, L. (2025). Generalizable neural-network parameterization of mesoscale eddies in idealized and global ocean models. *Geophysical Research Letters*, *52*(19), e2025GL117046. doi:https://doi.org/10.1029/2025GL117046

6. Zhang, C., **Perezhogin**, **P.**, Adcroft, A., & Zanna, L. (2025). Addressing out-of-sample issues in multi-layer convolutional neural-network parameterization of mesoscale eddies applied near coastlines. *Journal of Advances in Modeling Earth Systems (JAMES)*, *17*(5), e2024MS004819. doi:https://doi.org/10.1029/2024MS004819

7. Falasca, F., **Perezhogin**, **P.**, & Zanna, L. (2024). Data-driven dimensionality reduction and causal inference for spatiotemporal climate fields. *Phys. Rev. E*, *109*, 044202. doi:https://doi.org/10.1103/PhysRevE.109.044202

8. **Perezhogin**, **P.**, Zhang, C., Adcroft, A., Fernandez-Granda, C., & Zanna, L. (2024). A stable implementation of a data-driven scale-aware mesoscale parameterization. *JAMES*, *16*(10), e2023MS004104. doi:https://doi.org/10.1029/2023MS004104

9. **Perezhogin**, **P.**, & Glazunov, A. (2023). Subgrid Parameterizations of Ocean Mesoscale Eddies Based on Germano Decomposition. *JAMES*, *15*(10). doi:https://doi.org/10.1029/2023ms003771

10. **Perezhogin**, **P.**, Zanna, L., & Fernandez-Granda, C. (2023). Generative Data-Driven Approaches for Stochastic Subgrid Parameterizations in an Idealized Ocean Model. *JAMES*, *15*(10), e2023MS003681. doi:https://doi.org/10.1029/2023MS003681

11. Ross, A., Li, Z., **Perezhogin**, **P.**, Fernandez-Granda, C., & Zanna, L. (2023). Benchmarking of machine learning ocean subgrid parameterizations in an idealized model. *JAMES*, *15*(1), e2022MS003258. doi:https://doi.org/10.1029/2022MS003258

12. Zasko, G. V., Glazunov, A. V., Mortikov, E. V., Nechepurenko, Y. M., & **Perezhogin**, **P.** (2023). Optimal energy growth in stably stratified turbulent Couette flow. *Boundary-Layer Meteorology*, *187*(1-2), 395–421. doi:https://doi.org/10.1007/s10546-022-00744-3

13. Zhang, C., **Perezhogin**, **P.**, Gultekin, C., Adcroft, A., Fernandez-Granda, C., & Zanna, L. (2023). Implementation and Evaluation of a Machine Learned Mesoscale Eddy Parameterization Into a Numerical Ocean Circulation Model. *JAMES*, *15*(10), e2023MS003697. doi:https://doi.org/10.1029/2023MS003697

14. **Perezhogin**, **P.**, Chernov, I., & Iakovlev, N. (2021). Advanced parallel implementation of the coupled ocean–ice model FEMAO (version 2.0) with load balancing. *Geoscientific Model Development*, *14*(2), 843–857. doi:https://doi.org/10.5194/gmd-14-843-2021

15. **Perezhogin**, **P.** (2020a). 2D turbulence closures for the barotropic jet instability simulation. *Russian Journal of Numerical Analysis and Mathematical Modelling*, *35*(1), 21–35. doi:https://doi.org/10.1515/rnam-2020-0003

16. **Perezhogin**, **P.** (2020b). Testing of kinetic energy backscatter parameterizations in the NEMO ocean model. *Russian Journal of Numerical Analysis and Mathematical Modelling*, *35*(2), 69–82. doi:https://doi.org/10.1515/rnam-2020-0006

17. **Perezhogin**, **P.**, Glazunov, A. V., & Gritsun, A. S. (2019). Stochastic and deterministic kinetic energy backscatter parameterizations for simulation of the two-dimensional turbulence. *Russian Journal of Numerical Analysis and Mathematical Modelling*, *34*(4), 197–213. doi:https://doi.org/10.1515/rnam-2019-0017

18. Dymnikov, V., & **Perezhogin**, **P.** (2018). Systems of hydrodynamic type that approximate two-dimensional ideal fluid equations. *Izvestiya, Atmospheric and Oceanic Physics*, *54*, 232–241. doi:https://doi.org/10.1134/S0001433818030040

19. **Perezhogin**, **P.**, & Dymnikov, V. (2017). Modeling of quasi-equilibrium states of a two-dimensional ideal fluid. *Doklady Physics*, *62*, 248–252. doi:https://doi.org/10.1134/S1028335817050032

20. **Perezhogin**, **P.**, Glazunov, A. V., Mortikov, E. V., & Dymnikov, V. P. (2017). Comparison of numerical advection schemes in two-dimensional turbulence simulation. *Russian Journal of Numerical Analysis and Mathematical Modelling*, *32*(1), 47–60. doi:https://doi.org/10.1515/rnam-2017-0005

Conference Proceedings

1. **Perezhogin**, **P.**, Balakrishna, A., & Agrawal, R. (2024). Large eddy simulation of ocean mesoscale eddies. In *Center for Turbulence Research, Proceedings of the Summer Program*. Retrieved from https://web.stanford.edu/group/ctr/ctrsp24/vi07_PEREZHOGIN.pdf

2. Pedersen, C., Zanna, L., Bruna, J., & **Perezhogin**, **P.** (2023). Reliable coarse-grained turbulent simulations through combined offline learning and neural emulation. In *SynS & ML ICML*. doi:https://doi.org/10.48550/arXiv.2307.13144

Preprints

1. Shi, J.-R., **Perezhogin**, **P.**, Zanna, L., & Adcroft, A. (2026). *Impact of data-driven eddy parameterization on climate state in an idealized coupled cesm model*. doi:https://doi.org/10.48550/arXiv.2603.25843

2. Yuan, Y., Rusak, J., Merose, A., Subel, A., **Perezhogin**, **P.**, Adcroft, A., … Zanna, L. (2026). *Samudra 2: Scaling ocean emulators across resolutions*. doi:https://doi.org/10.48550/arXiv.2606.02610

3. Zanna, L., Gregory, W., **Perezhogin**, **P.**, Sane, A., Zhang, C., Adcroft, A. et al. (2025). *A Framework for Hybrid Physics-AI Coupled Ocean Models*. doi:https://doi.org/10.48550/arXiv.2510.22676

Note: *Additional* publications can be found at pperezhogin.github.io/publications

CV was created on October 25, 2026